

1. MC.09.02

Which of the following is part of the control process?

- ☐
- a. Monitoring of actual activity
- ☐
- b. Comparison of actual with planned activity
- ☐
- c. Investigating
- ☐
- d. Taking corrective action
- ☒
- e. All of these



Feedback

Check My Work

The control process consists of monitoring actual activity, comparing actual with planned outcomes, investigating unusual outcomes, and taking corrective action.

Solution
e

2. MC.09.03

Which of the following is *not* an advantage of budgeting?

- ☐
- a. It forces managers to plan.
- ☐
- b. It provides information for decision-making.
- ☒
- c. It guarantees an improvement in organizational efficiency.
- ☐
- d. It provides a standard for performance evaluation.
- ☐
- e. It improves communication and coordination.



Feedback

Check My Work

Advantages of budgeting are planning, information for decision makers, performance evaluation standards, and improved communication and coordination.

Solution
c

3. MC.09.04

The budget committee

- ☐
- a. reviews the budget.
- ☐
- b. resolves differences that arise as the budget is prepared.
- ☐
- c. approves the final budget.
- ☐
- d. is directed (typically) by the controller.
- ☒
- e. All of these.



Feedback

Check My Work

The budget committee reviews the budget, provides policy guidelines and budgetary goals, resolves differences that arise, approves final budget, and monitors actual performance.

Solution
e

4. MC.09.05

A moving, 12-month budget that is updated monthly is



- ☐ a. not used by manufacturing firms.
- ☐ b. a waste of time and effort.
- ☐ c. a master budget.
- ☒ d. a continuous budget.
- ☐ e. always used by firms that prepare a master budget.

Feedback

Check My Work

A continuous budget is a moving 12-month budget.

Solution
d

5. MC.09.06.ALGO (Algorithmic)

Which of the following is *not* part of the operating budget?



- ☐ a. The production budget
- ☒ b. The capital budget
- ☐ c. The direct labor budget
- ☐ d. The sales budget
- ☐ e. The selling and administrative expenses budget

Feedback

Check My Work

The operating budget consists of the budgeted income statement and all supporting budgets and schedules.

Solution
b

6. MC.09.08.ALGO (Algorithmic)

The first step in preparing the sales budget is to



- ☒ a. prepare a sales forecast.
- ☐ b. review the production budget carefully.
- ☐ c. talk with past customers.
- ☐ d. assess the desired ending inventory of finished goods.
- ☐ e. review the company's strategic plan.

Feedback

Check My Work

Review the preparation of the sales budget.

Solution
a

7. MC.09.10.ALGO (Algorithmic)

A company requires 100 pounds of plastic to meet the production needs of a small toy. It currently has 10 pounds of plastic inventory. The desired ending inventory of plastic is 30 pounds. How many pounds of plastic should be budgeted for purchasing during the coming period?



- ☒ a. 120 pounds

- ☐ b. 80 pounds
- ☐ c. 110 pounds
- ☐ d. 130 pounds
- ☐ e. None of these

Feedback

Check My Work
Correct
Post-Submission

$$100 + 30 - 10 = 120$$

Solution
a

8. MC.09.11

A company plans to sell 220 units. The selling price per unit is \$24. There are 50 units in beginning inventory, and the company would like to have 20 units in ending inventory. How many units should be produced for the coming period?

- ☐ a. 250
- ☐ b. 200
- ☐ c. 230
- ☐ d. 220
- ☒ e. None of these



Feedback

Check My Work
Correct
Post-Submission

$$220 + 20 - 50 = 190$$

Solution
e

9. MC.09.12

When a company amasses data during the preparation of the master budget, it is primarily doing so for which of the following purposes?

- ☐ a. Descriptive
- ☒ b. Predictive
- ☐ c. Prescriptive
- ☐ d. Diagnostic



Feedback

Check My Work
Correct.
Solution
b

10. MC.09.13

A company has the following collection pattern: month of sale, 40%; month following sale, 60%.

If credit sales for January and February are \$100,000 and \$200,000, respectively, the cash collections for February are?

- ☒ a. \$140,000.



- ☐ b. \$300,000.
- ☐ c. \$120,000.
- ☐ d. \$160,000.
- ☐ e. \$80,000.

Feedback

Check My Work
Correct
Post-Submission

$$(0.40 \times \$200,000) + (0.60 \times \$100,000) = \$140,000$$

Solution
a

11. MC.09.14

The percentage of accounts receivable that is uncollectible can be ignored for cash budgeting because

- ✓ ☒ a. no cash is received from an account that defaults.
- ☐ b. it is included in cash sales.
- ☐ c. it appears on the budgeted income statement.
- ☐ d. for most companies, it is not a material amount.
- ☐ e. None of these

Feedback

Check My Work

The amount a company deems for bad debt expense against receivables is not expected to be received in cash.

Solution
a

12. MC.09.15

Suppose that a factory increases training costs in order to reduce the waste of materials in production. That training cost is included in the

- ✓ ☒ d. the fixed factory overhead budget.
- ☐ a. sustainability budget.
- ☐ b. the variable selling expense budget.
- ☐ c. the direct materials purchases budget.
- ☐ e. the direct labor budget.

Feedback

Check My Work
Correct.
Solution
d

13. MC.09.16

Which of the following items is a possible example of myopic behavior?

- ☐ a. Failure to promote deserving employees
- ☐ b. Reducing expenditures on preventive maintenance
- ☐ c. Cutting back on new product development

-  ☐ d. Buying cheaper, lower-quality materials so that the company does not exceed the materials purchases budget.
- ☐ e. All of these

Feedback

Check My Work

Myopic behavior occurs when actions affect short-term performance, but at the expense of the long-term budget.

Solution
e

14. MC.09.17

For performance reporting, it is best to compare actual costs with budgeted costs using

-  ☐ static budgets.
- ☐ short-term budgets.
- ☐ master budgets.
- ☒ flexible budgets.
- ☐ None of these

Feedback

Check My Work

A budget with financial results for a range of activity levels is best for performance reporting.

Solution
d

15. MC.09.18

To create a meaningful performance report, actual costs and expected costs should be compared

-  ☐ hourly.
- ☐ weekly.
- ☒ at the actual level of activity.
- ☐ at the average level of activity.
- ☐ at the budgeted level of activity.

Feedback

Check My Work

A meaningful performance report uses an after-the-fact budget. Review how after-the-fact budgets are prepared.

Solution
c

16. MC.09.19

To help assess performance, managers should use a

- ☐ before-the-fact flexible budget.
- ☐ static budget.
- ☐ continuous budget.
- ☐ master budget.



None of these

Feedback

Check My Work

A meaningful performance report compares actual costs to budgeted costs at actual levels of production.

Solution
e

17. MC.09.20

A firm comparing the actual variable costs of producing 10,000 units with the total variable costs of a static budget based on 9,000 units would probably notice

- ☐ large favorable variances.
- ☐ no variances.
- ☐ small unfavorable variances.
- ☐ small favorable variances.
- ☒ large unfavorable variances.



Feedback

Check My Work

Decide if the output causes a variance assuming a static budget. If so, is the variance favorable or unfavorable, meaning more or less costly than budget? A variance of 10% or more is considered large.

Solution
e

18. MC.09.63

A company expects the following sales for the coming year:

	<u>1st Quarter</u>	<u>2nd Quarter</u>	<u>3rd Quarter</u>	<u>4th Quarter</u>
Units	40,000	60,000	80,000	100,000
Average selling price	\$6	\$6	\$6	\$8

Budgeted sales revenue for the year is:

- ☐ a. \$1,050,000.
- ☐ b. \$1,650,000.
- ☐ c. \$1,260,000.
- ☒ d. \$1,880,000.



Feedback

Check My Work
Correct
Post-Submission

Budgeted Sales Revenue = [(Expected Sales in Units for 1st Quarter + Expected Sales in Units for 2nd Quarter + Expected Sales in Units for 3rd Quarter) × Average Selling Price] + (Expected Sales in Units for 4th Quarter × Average Selling Price) = [(40,000 units + 60,000 units + 80,000 units) × \$6] + (100,000 units × \$8) = \$1,880,000

19. MC.09.67

Unique Company provided the following budgeted data for July:

Direct materials	\$70,000
Direct labor	\$45,000
Overhead	\$110,000
Beginning finished goods	\$20,000
Ending finished goods	\$35,000
Production in units	15,000

What is the budgeted cost of goods sold?

- 
- ☐

a. \$184,000
- ☐

b. \$165,000
- ☒

c. \$210,000
- ☐

d. \$179,000

Feedback

Check My Work
Correct
Post-Submission

Budgeted Manufacturing Costs = Direct Materials Used + Direct Labor Used + Overhead = \$70,000 + \$45,000 + \$110,000 = \$225,000
Budgeted Cost of Goods Sold = Budgeted Manufacturing Costs + Beginning Finished Goods – Ending Finished Goods = \$225,000 + \$20,000 – \$35,000 = \$210,000

20. MC.09.80

A company plans to sell 37,000 units of Product No. 99 in November, and each of these units requires five square feet of raw material. Pertinent data follow:

Product No. 99 Raw Materials		
Actual November 1 inventory	7,000	10,800 square feet
Estimated November 30 inventory	5,800	? square feet

If the company purchases 205,000 square feet of raw materials during the month, the estimated raw materials inventory on November 30 would be:

- 
- ☐

a. 42,800 square feet.
- ☐

b. 41,000 square feet.
- ☐

c. 53,000 square feet.
- ☒

d. 36,800 square feet.

Feedback

Check My Work
Correct

$37,000 - 7,000 + 5,800 = 35,800$ finished; $35,800 \times 5 = 179,000$ units of raw materials; $179,000 - 10,800 = 168,200$ units of raw materials; $205,000 - 168,200 = 36,800$ sq. ft. at November 30.

21. MC.09.85

A company produces two products: SR200 and TX500. Budgeted sales for four months are as follows:

SR200 TX500

May	8,000	20,000
June	13,000	32,000
July	11,000	39,000
August	18,000	46,000

The company's ending inventory policy is that SR200 should have 15% of next month's sales in ending inventory and TX500 should have 20% of next month's sales in ending inventory. On May 1, there were 1,500 units of SR200 and 8,000 units of TX500.

TX500 requires six units of Component A. (SR200 does not use Component A.) There were 25,000 units of Component A in inventory on May 1. Kenner wants to have 20% of the following month's production needs in inventory for Component A. What is the desired ending inventory of Component A for May?

- ☐ a. 25,000
- ☐ b. 133,400
- ☒ c. 40,800
- ☐ d. 125,480



Feedback

Check My Work
Correct
Post-Submission

	<u>May</u>	<u>June</u>
Production	18,400	33,400
× Units of Component A	<u> × 6 </u>	<u> × 6 </u>
Component A needed for production	110,400	<u>200,400</u>
Desired ending inventory*	<u>40,080</u>	
Units needed	158,400	
Less: Beginning inventory	<u>(25,000)</u>	
Production	<u><u>133,400</u></u>	

*(33,400 × 6) × 20% = 40,080

22. MC.09.89

A company plans to sell 142,000 units in November and 178,000 units in December. The company's policy is that 10% of the following month's sales must be in ending inventory. On November 1, there were 14,000 units in inventory.

It takes 30 minutes of direct labor time to make one unit. Direct labor wages average \$18 per hour. Variable overhead is applied at the rate of \$5 per direct labor hour. Fixed overhead is budgeted at \$57,000 per month. What is the direct labor cost budgeted for November?

- ☐ a. \$950,600
- ☐ b. \$1,181,500

- ☐ c. \$2,152,000
☒ d. \$1,312,200

Feedback

Check My Work
 Correct
 Post-Submission

November Production = $142,000 + (0.10 \times 178,000) - 14,000 = 145,800$ units
 Direct Labor Hours = $0.50 \text{ hour per unit} \times 145,800 = 72,900$ hours
 Direct Labor Cost = $\$18 \times 72,900 = \$1,312,200$

23. MC.09.93

A company provided the following information for last month:

Production in units	5,000
Direct materials cost	\$8,000
Direct labor cost	\$12,000
Overhead cost	\$9,600
Sales commission per unit sold	\$5
Price per unit sold	\$30
Fixed selling and administrative expense	\$8,000

There were no beginning and ending inventories. What is the gross margin for the company last month?

- ☐ a. \$80,400
☐ b. \$60,500
☐ c. \$105,600
☒ d. \$120,400

Feedback

Check My Work
 Correct
 Post-Submission

Sales ($\$30 \times 5,000$)	\$87,000
Cost of goods sold ($\$8,000 + \$12,000 + \$9,600$)	<u>29,600</u>
Gross margin	<u>\$120,400</u>

24. MC.09.102

A sporting goods store sells bicycles throughout the southeastern United States. The following data were taken from the most recent quarterly sales forecast:

	End-of-Month	
	Expected Sales	Target Inventory
July	1,990 units	400 units
August	2,150 units	490 units
September	2,070 units	460 units

On the basis of the information presented, how many bicycles should the company purchase in August?



- ☐ a. 2,330
- ☐ b. 2,630
- ☒ c. 2,240
- ☐ d. 2,040

Feedback

Check My Work
Correct
Post-Submission

$(2,150 + 490) - 400 = 2,240$ units purchased in August

25. MC.09.112

A company makes all sales on account, subject to the following collection pattern: 30% are collected in the month of sale; 60% are collected in the first month after sale; and 10% are collected in the second month after sale. If sales for October, November, and December were \$78,000, \$68,000, and \$48,000, respectively, what was the budgeted receivables balance on December 31?



- ☒ a. \$40,400
- ☐ b. \$56,000
- ☐ c. \$37,400
- ☐ d. \$42,600

Feedback

Check My Work
Correct
Post-Submission

$\text{Receivables} - \text{Collected} = (\$78,000 + \$68,000 + \$48,000) - [(\$78,000 \times 100\%) + (\$68,000 \times 90\%) + (\$48,000 \times 30\%)] =$
 $\$194,000 - \$153,600 = \$40,400$

26. MC.09.128

A company makes all its sales on account. The company's accounts receivable payment experience is as follows:

Percent paid in the month of sale	30%
Percent paid in the month after the sale	65%
Percent paid in the second month after the sale	3%

The company provided the following information on sales:

September	\$110,000
October	120,000
November	230,000
December (expected)	260,000

What are the expected cash receipts in the month of November?



- ☐ a. \$40,200
- ☐ b. \$190,000
- ☒ c. \$150,300
- ☐ d. \$160,600

Check My Work
Correct
Post-Submission

Expected Cash Receipts in November = (Sales Made in November × Percent Paid in Month of Sale) + (Sales Made in October × Percent Paid in Month after Sale) + (Sales Made in September × Percent Paid in Second Month after Sale) = $(\$230,000 \times 0.30) + (\$120,000 \times 0.65) + (\$110,000 \times 0.03) = \$69,000 + \$78,000 + \$3,300 = \$150,300$

27. MC.09.144

Jason Inc. produces leather purses. Jason has developed a static budget for the first quarter based on 20,000 direct labor hours. During the quarter, the actual activity was 22,000 direct labor hours. Data for the first quarter are summarized as follows:

	<u>Static Budget</u> <u>(20,000 hours)</u>	<u>Actual Costs</u> <u>(22,000 hours)</u>
Direct materials cost	\$ 80,000	\$ 87,000
Direct labor cost	160,000	174,000
Building rental	<u>48,000</u>	<u>50,000</u>
Total	<u>\$288,000</u>	<u>\$311,000</u>

What is the flexible budget variance for the first quarter?



- ☐ a. \$23,000 U
- ☒ b. \$1,000 F
- ☐ c. \$23,000 F
- ☐ d. \$1,000 U

Check My Work
Correct
Post-Submission

	<u>Flexible Budget</u> <u>(22,000 hours)</u>	<u>Actual Costs</u> <u>(22,000 hours)</u>	<u>Variance</u>
Direct materials	\$ 88,000	\$ 87,000	\$1,000 F
Direct labor	176,000	174,000	2,000 F
Building rental	<u>48,000</u>	<u>50,000</u>	<u>2,000 U</u>
Total	<u>\$312,000</u>	<u>\$311,000</u>	<u>\$1,000 F</u>

The flexible budget here is created by creating standard rates from the static budget. \$80,000 of direct labor divided by 20,000 labor hours yields \$4 per labor hour.

This standard rate is multiplied by the actual labor hours to get the standard labor hours for the actual activity level.

28. PR.09.221.ALGO (Algorithmic)

Bottleneck Company has put together the following data in order to complete its operating budget for the second quarter in the current year:

April May June July

Sales (units) 74,900 68,800 63,200 67,200

Additional information:

Company policy requires that 62% of next month's sales (in units) be in ending inventory. This policy was met in March.

It takes 1 hours of direct labor to produce one unit.

The average wage cost is \$20.
Variable overhead rate is \$6 per direct labor hour, and fixed overhead is \$17,500 per month.

Required:

A. Prepare a production budget for April, May, June and the quarter in total.

Bottleneck Company Production Budget For the Quarter Ended June 30				
	April	May	June	Total
	✓	✓	✓	✓
Sales in units	74900	68800	63200	206900
	✓	✓	✓	✓
Desired ending inventory	42656	39184	41664	41664
	✓	✓	✓	✓
Total needs	117556	107984	104864	248564
	✓	✓	✓	✓
Less: Beginning inventory	46438	42656	39184	46438
	✓	✓	✓	✓
Units to be produced	71118	65328	65680	202126

Feedback

Check My Work

Correct

Post-Submission

A.

Desired ending inventory:

April (68,800 × 0.62) = 42,656

May (63,200 × 0.62) = 39,184

June (67,200 × 0.62) = 41,664

Less: Beginning inventory:

April (74,900 × 0.62) = 46,438

May (68,800 × 0.62) = 42,656

June (63,200 × 0.62) = 39,184

B. Prepare a direct labor budget for April, May, June and the quarter in total.

Bottleneck Company Direct Labor Budget For the Quarter Ended June 30				
	April	May	June	Total
	✓	✓	✓	✓
Units to be produced	71118	65328	65680	202126
	✓	✓	✓	✓
Direct labor hours per unit	1	1	1	1
	✓	✓	✓	✓
Total hours needed	71118	65328	65680	202126

	\$	✓	\$	✓	\$	✓	\$	✓
Average wage rate per hour	20		20		20		20	
	\$	✓	\$	✓	\$	✓	\$	✓
Total direct labor cost	1422360		1306560		1313600		4042520	

Feedback

Check My Work
Correct

C. Prepare an overhead budget for April, May, June and the quarter in total.

Bottleneck Company
Overhead Budget
For the Quarter Ended June 30

	April	May	June	Total
	✓	✓	✓	✓
Budgeted direct labor hours	71118	65328	65680	202126
	\$	\$	\$	\$
Variable overhead rate	6	6	6	6
	\$	\$	\$	\$
Budgeted variable overhead	426708	391968	394080	1212756
	✓	✓	✓	✓
Budgeted fixed overhead	17500	17500	17500	52500
	\$	\$	\$	\$
Total overhead	444208	409468	411580	1265256

Feedback

Check My Work
Correct

Feedback

Check My Work
Correct
Solution

A. Prepare a production budget for April, May, June and the quarter in total.

Bottleneck Company
Production Budget
For the Quarter Ended June 30

	April	May	June	Total
1 Sales in units	74900	68800	63200	206900
Desired ending inventory	42656	39184	41664	41664
Total needs	117556	107984	104864	248564
Less: Beginning inventory	46438	42656	39184	46438
Units to be produced	71118	65328	65680	202126

2

B. Prepare a direct labor budget for April, May, June and the quarter in total.

Bottleneck Company
Direct Labor Budget
For the Quarter Ended June 30

	April	May	June	Total
Units to be produced	71118	65328	65680	202126
Direct labor hours per unit	1	1	1	1
Total hours needed	71118	65328	65680	202126
Average wage rate per hour	\$ 20	\$ 20	\$ 20	\$ 20
Total direct labor cost	\$ 1422360	\$ 1306560	\$ 1313600	\$ 4042520

C. Prepare an overhead budget for April, May, June and the quarter in total.

Bottleneck Company
Overhead Budget
For the Quarter Ended June 30

	April	May	June	Total
Budgeted direct labor hours	71118	65328	65680	202126
Variable overhead rate	\$ 6	\$ 6	\$ 6	\$ 6
Budgeted variable overhead	\$ 426708	\$ 391968	\$ 394080	\$ 1212756
Budgeted fixed overhead	17500	17500	17500	52500
Total overhead	\$ 444208	\$ 409468	\$ 411580	\$ 1265256

29. PR.09.226.ALGO (Algorithmic)

HairMaster Salons anticipates giving 120 permanents in May, 100 in June, and 120 in July. HairMaster needs one permanent wave kit for each perm, along with two boxes of wave tissues. Its inventory policy is to have 10% of the following month's materials needs on hand. On May 1, there were 15 wave kits and 4 boxes of wave tissues on hand. **(Round any fractions of a unit to the nearest whole unit.)**

A. What is the number of wave kits to be purchased in May?

✓

115
kits

B. What is the number of wave kits to be purchased in June?

✓

102
kits

C. What is the number of boxes of tissues to be purchased in May?

✓

256
boxes

D. What is the number of boxes of tissues to be purchased in June?

✓

204
boxes

Feedback

Check My Work
Correct
Post-Submission

A.

May Wave Kit Purchases

$$= 120 + 10 - 15 = 115$$

B.

June Wave Kit Purchases

$$= 100 + 12 - 10 = 102$$

C.

May Purchases of Tissue Boxes

$$= (120 \times 2 \text{ boxes}) + (100 \times 2 \text{ boxes} \times 0.10) - 4 = 256$$

D.

June Purchases of Tissue Boxes

$$= (100 \times 2 \text{ boxes}) + (120 \times 2 \text{ boxes} \times 0.10) - (100 \times 2 \text{ boxes} \times 0.10) = 204$$

Solution

HairMaster Salons anticipates giving 120 permanents in May, 100 in June, and 120 in July. HairMaster needs one permanent wave kit for each perm, along with two boxes of wave tissues. Its inventory policy is to have 10% of the following month's materials needs on hand. On May 1, there were 15 wave kits and 4 boxes of wave tissues on hand. **(Round any fractions of a unit to the nearest whole unit.)**

A. What is the number of wave kits to be purchased in May?

115 kits

B. What is the number of wave kits to be purchased in June?

102 kits

C. What is the number of boxes of tissues to be purchased in May?

256 boxes

D. What is the number of boxes of tissues to be purchased in June?

204 boxes

30. PR.09.227.ALGO (Algorithmic)

Foster Company makes power tools. The sales budget for drills for the first four months of the year is as follows:

Month	Unit Sales
January	26,000
February	12,000
March	24,000
April	27,000

Foster has taken a just-in-time approach to production and wants only 8% of the next month's sales needs in ending inventory. January 1 inventory of drills was zero. Each drill takes 15 minutes of direct labor at \$20 per hour. The factory overhead formula is \$26,000 + \$1.70 per direct labor hour.

Required: Round your answers to the nearest whole unit or whole dollar.

A. What is the budgeted production for January?



26960

drills

B. What is the budgeted production for February?



12960

drills

C. What is the budgeted production for the entire first quarter of the year?



64160

drills

D. What is the budgeted direct labor cost for January?

\$

134800

E. What is the budgeted direct labor cost for February?

\$

64800

F. What is the budgeted variable overhead for March?

\$

10302

G. What is the budgeted total overhead for March?

\$

36302

Feedback

Check My Work

Correct

Post-Submission

A.

Budgeted Production for January

$$26,000 + 960 - 0 = 26,960$$

B.

Budgeted Production for February

$$12,000 + 1,920 - 960 = 12,960$$

C.

Budgeted Production for Entire First Quarter of Year

$$26,960 + 12,960 + 24,240^* = 64,160$$

$$^*24,000 + 2,160 - 1,920$$

D.

Budgeted Direct Labor Cost for January

$$26,960 \times 0.25^* \times \$20 = \$134,800$$

$$^*15 \div 60$$

E.

Budgeted Direct Labor Cost for February

$$12,960 \times 0.25 \times \$20 = \$64,800$$

F.

Budgeted Variable Overhead for March

$$24,240 \times 0.25 \times \$1.70 = \$10,302$$

G.

March Budgeted Total Overhead

$$\$26,000 + \$10,302 = \$36,302$$

Solution

Foster Company makes power tools. The sales budget for drills for the first four months of the year is as follows:

Month	Unit Sales
--------------	-------------------

January	26,000
---------	--------

February	12,000
----------	--------

March	24,000
-------	--------

April	27,000
-------	--------

Foster has taken a just-in-time approach to production and wants only 8% of the next month's sales needs in ending inventory. January 1 inventory of drills was zero. Each drill takes 15 minutes of direct labor at \$20 per hour. The factory overhead formula is $\$26,000 + \1.70 per direct labor hour.

Required: Round your answers to the nearest whole unit or whole dollar.

A. What is the budgeted production for January?

drills

B. What is the budgeted production for February?

drills

C. What is the budgeted production for the entire first quarter of the year?

drills

D. What is the budgeted direct labor cost for January?

\$

E. What is the budgeted direct labor cost for February?

\$

F. What is the budgeted variable overhead for March?

\$

G. What is the budgeted total overhead for March?

\$

31. PR.09.230.ALGO (Algorithmic)

You have decided to throw a party next weekend for 19 friends. The friends are going to bring healthy food, so all you need to do is buy the drinks. You estimate that, on average, each person will drink 8 bottles of soft drinks. Three of your friends will drink only natural soda without unneeded color, so Sulo Ginger Ale should work well for them. For the others and yourself, you decide to buy Sulo Cola. Before going online, you check the refrigerator—you already have 7 bottles of Sulo Ginger Ale and 11 of Sulo Cola. Since this is the end of the semester, you decide that you don't really want any of the soft drinks on hand after the party. Now, you are ordering on the Internet.

A. How many bottles of Sulo Ginger Ale do you plan to buy?



17
bottles

B. How many bottles of Sulo Cola do you plan to buy?



125
bottles

Feedback

Check My Work
Correct
Post-Submission

Ginger Ale Cola

To drink at the party:		
3 friends × 8 bottles	24	
17 friends × 8 bottles	136	
+ Desired ending inventory	0	0
Bottles needed	24	136
– Beginning inventory	7	11
Bottles to purchase	17	125

Solution

You have decided to throw a party next weekend for 19 friends. The friends are going to bring healthy food, so all you need to do is buy the drinks. You estimate that, on average, each person will drink 8 bottles of soft drinks. Three of your friends will drink only natural soda without unneeded color, so Sulo Ginger Ale should work well for them. For the others and yourself, you decide to buy Sulo Cola. Before going online, you check the refrigerator—you already have 7 bottles of Sulo Ginger Ale and 11 of Sulo Cola. Since this is the end of the semester, you decide that you don't really want any of the soft drinks on hand after the party. Now, you are ordering on the Internet.

A. How many bottles of Sulo Ginger Ale do you plan to buy?

bottles

B. How many bottles of Sulo Cola do you plan to buy?

bottles

32. PR.09.232.ALGO (Algorithmic)

Rodelli Company usually sells about 20% of its merchandise during a month for cash with the remaining sales on account. The company's accounts receivable payment history is as follows: 30% in the month of sale, 50% in the month following, and 15% in the second month following the sale. Total budgeted sales for the second quarter are as follows:

April \$122,000
May 141,000
June 74,000

A. What are the expected cash sales in June?

\$

14800

B. What are the expected receipts in June from accounts receivable for sales made in April?

\$

14640

C. What are the expected receipts in June from accounts receivable for sales made in May?

\$

56400

D. What are the total expected cash receipts in June?

\$

103600

E. From the given accounts receivable history information, receipts from accounts receivable do not equal 100%. Does this amount appear on the cash budget?

No

Feedback

Check My Work
Correct
Post-Submission

A.

June Cash Sales = \$74,000 × 0.20 = \$14,800

B.

Receipts on Accounts Receivable for Sales Made in April = 0.80 × \$122,000 × 0.15 = \$14,640

C.

Receipts on Accounts Receivable for Sales Made in May = 0.80 × \$141,000 × 0.50 = \$56,400

D.

Total Cash Expected in June = \$14,800 + \$14,640 + \$56,400 + (0.80 × \$74,000 × 0.30) = \$103,600

E.

30% + 50% + 15% = 95%

The remaining 5% is uncollectible. It does not appear on the cash budget because there is no cash involved.

Solution

Rodelli Company usually sells about 20% of its merchandise during a month for cash with the remaining sales on account. The company's accounts receivable payment history is as follows: 30% in the month of sale, 50% in the month following, and 15% in the second month following the sale. Total budgeted sales for the second quarter are as follows:

April \$122,000

May 141,000

June 74,000

A. What are the expected cash sales in June?

\$

B. What are the expected receipts in June from accounts receivable for sales made in April?

\$

C. What are the expected receipts in June from accounts receivable for sales made in May?

\$

D. What are the total expected cash receipts in June?

\$ 103600

E. From the given accounts receivable history information, receipts from accounts receivable do not equal 100%. Does this amount appear on the cash budget?

No

33. PR.09.238.ALGO (Algorithmic)

It is May 28 and you were just hired for a summer job that will pay you (net of taxes) \$960 per month. You start on June 1 and will work until school starts—halfway through August. Your scholarship pays for tuition, room, and board. But you must buy books; pay for transportation to and from school; and pay for clothing, any extra meals, entertainment, and so on. You have gathered the following data:

1. One round-trip airline ticket is \$200, and you'd like to come home for Thanksgiving. (Your parents will drive you there in August, and you will try to catch a ride home with another student in December).
2. Books are estimated to cost about \$540 per semester for your anticipated major.
3. Supplies should be another \$130.
4. Clothing might run \$140—you already have almost everything you think you'll need.
5. There are 16 weeks in the semester, and you think you'll need \$42 per week for allowance to cover extra meals and entertainment.
6. Before school even starts, you need to cover any summer expenses, including going out with friends. \$20 a week sounds about right since all your friends will be working and saving for college as well. There are 11 weeks of summer.

Right now, you have \$290 in your checking account.

Prepare a cash budget for the summer and the first semester of college. (Do the entire time period; do not break it down by week or by month.)

Cash Budget

Line Item Description	Amount
Beginning balance checking account	\$ 290
Salary from summer job	2400
Cash available	\$ 2690
Less: Disbursements	
Airline ticket	200
Books	540
Supplies	130
Clothing	140
Weekly allowance at school	672

Weekly allowance during the summer	✓	220
Total disbursements	✓	\$ 1902
Estimated ending balance		\$ 788

Feedback

Check My Work
Correct
Post-Submission

Salary from summer job ($2.5 \times \$960$) = \$2,400
 Weekly allowance at school ($16 \times \$42$) = \$672
 Weekly allowance during the summer ($11 \times \$20$) = \$220

Solution

It is May 28 and you were just hired for a summer job that will pay you (net of taxes) \$960 per month. You start on June 1 and will work until school starts—halfway through August. Your scholarship pays for tuition, room, and board. But you must buy books; pay for transportation to and from school; and pay for clothing, any extra meals, entertainment, and so on. You have gathered the following data:

1. One round-trip airline ticket is \$200, and you'd like to come home for Thanksgiving. (Your parents will drive you there in August, and you will try to catch a ride home with another student in December).
2. Books are estimated to cost about \$540 per semester for your anticipated major.
3. Supplies should be another \$130.
4. Clothing might run \$140—you already have almost everything you think you'll need.
5. There are 16 weeks in the semester, and you think you'll need \$42 per week for allowance to cover extra meals and entertainment.
6. Before school even starts, you need to cover any summer expenses, including going out with friends. \$20 a week sounds about right since all your friends will be working and saving for college as well. There are 11 weeks of summer.

Right now, you have \$290 in your checking account.

Prepare a cash budget for the summer and the first semester of college. (Do the entire time period; do not break it down by week or by month.)

Cash Budget

Line Item Description	Amount
Beginning balance checking account	\$ 290
Salary from summer job	2400
Cash available	\$ 2690
Less: Disbursements	
Airline ticket	200
Books	540
Supplies	130
Clothing	140
Weekly allowance at school	672
Weekly allowance during the summer	220
Total disbursements	\$ 1902

Estimated ending balance

\$ 788

34. PR.09.243

Larry Miller, the controller for Kipling Company, has been instructed to develop a flexible budget for overhead costs. The company produces two types of frozen desserts: icy and tasty. The two desserts use common raw materials in different proportions. The company expects to produce 200,000 gallons of each product during the coming year. Icy requires 0.25 direct labor hour per gallon, and Tasty requires 0.30. Larry has developed the following fixed and variable costs for each of the four overhead items:

Overhead Item Fixed Cost Variable Rate per DLH

Maintenance	\$52,000	\$1.20
Power		1.50
Indirect labor	79,500	4.80
Rent	54,000	

Required:

- a. Prepare an overhead budget for the expected activity level for the coming year.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	110000 hrs	✓
Variable costs:			
Maintenance	\$1.20	\$ 132000	✓
Power	1.50	165000	✓
Indirect labor	4.80	528000	✓
Total variable costs		\$ 825000	✓
Fixed costs:			
Maintenance		\$ 52000	✓
Indirect labor		79500	✓
Rent		54000	✓
Total fixed costs		\$ 185500	✓
Total overhead costs		\$ 1010500	✓

Feedback

Check My Work

a.

Activity Level*

*Icey:	$(0.25 \times 200,000)$	50,000
Tasty:	$(0.30 \times 200,000)$	60,000
Total DLH		<u>1,10,000</u>

b. Prepare an overhead budget that reflects production that is 10% higher than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	121000 hrs	✓
Variable costs:			
Maintenance	\$1.20	145200	\$ ✓
Power	1.50	181500	✓
Indirect labor	4.80	580800	✓
Total variable costs		907500	\$ ✓
Fixed costs:			
Maintenance		52000	\$ ✓
Indirect labor		79500	✓
Rent		54000	✓
Total fixed costs		185500	✓
Total overhead costs		1093000	\$ ✓

Feedback

b.
121,000 Hours*
*110,000 DLH × 110% = 121,000

c. Prepare an overhead budget that reflects production that is 10% lower than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company Overhead Budget For the Coming Year		
	Formula	99000 hrs
Variable costs:		
Maintenance	\$1.20	\$ 118800
Power	1.50	148500
Indirect labor	4.80	475200
Total variable costs		\$ 742500
Fixed costs:		
Maintenance		\$ 52000
Indirect labor		79500
Rent		54000
Total fixed costs		185500
Total overhead costs		\$ 928000

Feedback

Check My Work
Correct
Post-Submission

c.
99,000 Hours*
*110,000 DLH × 90% = 99,000

Feedback

Check My Work
Correct

- a. Prepare an overhead budget for the expected activity level for the coming year.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	110000 hrs
Variable costs:		
Maintenance	\$1.20	\$ 132000
Power	1.50	165000
Indirect labor	4.80	528000
Total variable costs		\$ 825000
Fixed costs:		
Maintenance		\$ 52000
Indirect labor		79500
Rent		54000
Total fixed costs		\$ 185500
Total overhead costs		\$ 1010500

- b. Prepare an overhead budget that reflects production that is 10% higher than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company
Overhead Budget
For the Coming Year

	Formula	121000 hrs
Variable costs:		
Maintenance	\$1.20	\$ 145200
Power	1.50	181500
Indirect labor	4.80	580800
Total variable costs		\$ 907500
Fixed costs:		
Maintenance		\$ 52000
Indirect labor		79500
Rent		54000
Total fixed costs		185500
Total overhead costs		\$ 1093000

- c. Prepare an overhead budget that reflects production that is 10% lower than expected (for both products). Assume this quantity is within the relevant range.

Kipling Company
Overhead Budget
For the Coming Year

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	Formula	99000	hrs
Variable costs:			
Maintenance	\$1.20	\$ 118800	
Power	1.50	148500	
Indirect labor	4.80	475200	
Total variable costs		\$ 742500	
Fixed costs:			
Maintenance		\$ 52000	
Indirect labor		79500	
Rent		54000	
Total fixed costs		185500	
Total overhead costs		\$ 928000	